SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I V.Humair(192524179) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Understanding of Concepts	25%	2	1	1	1
Explanation	25%	2	2	1	1
Application	20%	1	1	1	2
Organization	15%	1	1	2	1
Accuracy	10%	1	1	1	2
Clarity	5%	1	2	2	1

Total Marks 33 **/40**

Signature of the Faculty

Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

7.An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests of

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CO1 AT-1

Course :- Computer Networks

Course Code:- CSA 0709

Class Test- 33/40

① A Company needs to transfer a 150 MB design file from a client computer to a remote server over a network 1500-byte segments while the Data Link layer encapsulates by 40 bytes the total transmision time Data Link layers ensure reliable commuction

② Transport Layer protocal users a sliding window size of 6 frames Each frame carries 1024 bytes of user data. 4 frames header window

③ A 12,000-byte ip packet is transmitted through a router where the maximum Transmision 1500byte Assuming each fragment requires a 20-byte Ip head

④ A Transport layer protocal users a sliding window size of 6 frames each frame carries 1024 bytes. of users data if data 48 frames Calculate

file Transfer using Transport Layer and data Link Layer

Introduction:-

In Computer network, data is transferred from one Computer to another through different layers of the network model

given data:-

file size = 150mB

Network bandwith = 50 mbps

Segment size = 1500 bytes

Data Link layer header and trailer = 40 bytes per frame

we need to calculate the total number of segments and frames, transmission overhead, and total transmission time

1 mB = 10^6 bytes

150mB = 150×10^6

file size = 150,000,000 bytes

Each segment carries 1500 bytes

the number of segments is calculate using

$$\text{Number of segments} = \frac{\text{Total file size}}{\text{Segment size}}$$

$$= \frac{150,000,000}{1500}$$

therefore the Transport layer divides the file into 100;00 segments

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② given data

window size = 6 frames

Total frames = 48

Data per frame = 1024 bytes

one lost frame = per window

formula:

Number of windows = Total frames / window size

Calculation:-

$= \frac{48}{6}$

= 8 windows

Calculate the total data transmitted

formula:

Total data = number of frames x Data per from

Calculation:

48 x 1024

= 49,152 bytes

= 48 kB

Calculate the total number of retransmission

Number of windows = 8

Total retransmissions = 8 x 1

= 8 retransmision.

4. Explain flow Control:

flow control prevent a fast sender from sending more data then the receiver can handle.

Advantages:

① prevents receiver buffer overflow

② Controls the rate of data transmissions

③ Reduces unesary data loss

④ improves Communications efficency

⑤ Allows multiple frames to be transmitted before wating.

Number of windows = 8

Total data = 49,152 bytes

Total retransmission = 8 frames

(3)(3) given data:-

Total IP packet size = 12,000 bytes

MTU = 1500 bytes

IP header size = 20 bytes

Assume the 12,000 byte packet includes its original IP header.

i) Calculate the maximum data per fragment

formula:-

maximum fragment data = MTU - IP header

Calculation:-

1500 - 20

= 1480 bytes

(2) Calculate the original data size

Original data size = Total packet size - original IP header

= 12,000 - 20

= 11,980 bytes

(3) Calculate the number of fragments.

11,980 / 1480 = 8.09

∴ we need a fragment

= 8 × 1480

= 11,840 bytes

= 11,980 − 11,840

= 140 byte

Calculate the total header overhead:-

formula:

Total header overhead = number of fragments header size

Calculation:-

a × 20

= 180 bytes

Number of fragments = 9

Total header overhead = 180 bytes

Total transmitted size = 12,160 bytes

④ Sliding window protocal calculation and Explanation

given

- sliding window size: 6 frames
  Data carried by each frame: 1024 bytes
  Total frame transmitted = 48
  one frame in every window is lost and must be retransmited

1. Number of transmission windows Required

the number of window is calculated by dividing the total number of frames by the window size

$$\text{windows Required} : \frac{48}{6} = 8$$

∴ 8 transmision windows are Required

2. Number of Retransmisions

One frame is lost in Every window and must be retramitted

$$\text{Retransmision} - 8 \times 1 = 8$$

3. Total data Transmitted
Each frame carries 1024 bytes of user data

$$\text{Original Data} = 48 \times 1024$$

$$= 49,152 \text{ bytes}$$

⑤ increases throughput :- it helps transfer data faster and improves overall network performances

Conclusion :- the sender requires 8 transmission windows and 8 retransmissions the total data transmitted including retransmission is 57,344 bytes, the sliding window protocol improves communication Efficiency throug Effective bandwith utilization

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⑤ given data :-

Total file size = 600 mB

Check point interval = 60 mB

Data transferred before failure = 390 mB

1) calculate the number of Check points

formula :-

number of checkpoints = Data transferral

Calculation :-

390/60

= 6.5

six Complete checkpoints have been before the failure

Calculate the data to be retransmitted

Data transfered after the lastes Checkpoint

= 390 - 360

= 30 mB

③ Explan the improtance of synchroizate

① synchronization crates checkpoints during data transmission

② checkpoints help Communication resume after a failure

③ the system does not need to restart the entire transfer

- Number of Completed Chead Checkpoint = 6

latest CheckPant : 360 mB

Data to be transmitted = 30 mB

3 MB each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

8.A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

9.A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

10.A hospital transfers a patient's medical record of **240 MB** through an IoT-enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation

Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organizatio n	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professiona l presentatio n	Understanda ble with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand